

Seborrhoeic keratosis

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What is a seborrhoeic keratosis?

Seborrhoeic keratosis is a harmless warty spot that appears during adult life as a common sign of [skin ageing](#). Some people have hundreds of them.

Seborrhoeic keratosis (American spelling – seborrheic keratosis) is also called SK, basal cell papilloma, senile wart, brown wart, wisdom wart, or barnacle. The descriptive term, benign keratosis, is a broader term that is used to include the following related scaly skin lesions:

Seborrhoeic keratosis

[Solar lentigo](#) (which can be difficult to distinguish from a flat seborrhoeic keratosis)

[Lichen planus-like keratosis](#) (which arises from a seborrhoeic keratosis or a solar lentigo).



Seborrheic keratosis



Seborrheic keratosis



Pigmented seborrheic keratosis

Who gets seborrheic keratoses?

Seborrheic keratoses are extremely common. It has been estimated that over 90% of adults over the age of 60 years have one or more of them. They occur in males and females of all races, typically beginning to erupt in the 30s or 40s. They are uncommon under the age of 20 years.

What causes seborrheic keratoses?

The precise cause of seborrheic keratoses is not known.

The name is misleading, because they are not limited to a seborrheic distribution (scalp, mid-face, chest, upper back) as in [seborrheic dermatitis](#), nor are they formed from sebaceous glands, as is the case with [sebaceous hyperplasia](#), nor are they associated with sebum — which is greasy.

Seborrheic keratoses are considered degenerative in nature. As time goes by, seborrheic keratoses become more numerous. Some people inherit a tendency to develop a very large number of them.

Researchers have noted:

- Eruptive seborrheic keratoses can follow [sunburn](#) or [dermatitis](#).

- Skin friction may be the reason they appear in body folds.

- [Viral](#) cause (eg human papillomavirus) seems unlikely.

- Stable and clonal mutations or activation of FRFR3, PIK3CA, RAS, AKT1 and EGFR genes are found in seborrheic keratoses.

- Seborrheic keratosis can arise from [solar lentigo](#).

FRFR3 mutations also arise in solar lentigines. These mutations are associated with increased age and location on the head and neck, suggesting a role of ultraviolet radiation in these lesions.

Seborrhoeic keratoses do not harbour tumour suppressor gene mutations.

[Epidermal growth factor receptor inhibitors](#) (used to treat cancer) often result in an increase in verrucal (warty) keratoses.

What are the clinical features of seborrhoeic keratoses?

Seborrhoeic keratoses can arise on any area of skin, covered or uncovered, with the exception of palms and soles. They do not arise from mucous membranes.

Seborrhoeic keratoses have a highly variable appearance.

Flat or raised papule or plaque

1 mm to several cm in diameter

Skin coloured, yellow, grey, light brown, dark brown, black or mixed colours

Smooth, waxy or warty surface

Solitary or grouped in certain areas, such as within the scalp, under the breasts, over the spine or in the groin

They appear to stick on to the skin surface like barnacles.



Seborrhoeic keratosis



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[See more images of seborrhoeic keratoses](#)

[See dermoscopy images of seborrhoeic keratoses](#)

Variants of seborrheic keratoses

Variants of seborrheic keratoses include:

Solar lentigo: flat circumscribed pigmented patches in sun-exposed sites

Dermatosis papulosa nigra: small, pedunculated and heavily pigmented seborrheic keratoses on head and neck of darker-skinned individuals

Stucco keratoses: grey, white or yellow papules on the lower extremities

Inverted follicular keratosis

Large cell acanthoma

Lichenoid keratosis: an inflammatory phase preceding involution of some seborrheic keratoses and solar lentigines.



Florid lesions of stucco keratoses on the ankle



Dermatosis papulosa nigra



Irritated seborrheic keratosis



Complications of seborrheic keratoses

Seborrheic keratoses are not premalignant tumours. However:

Skin cancers are sometimes difficult to tell apart from seborrheic keratoses.

Skin cancer may by chance arise within or collide with a seborrheic keratosis.

Very rarely, eruptive seborrheic keratoses may denote an underlying internal malignancy, most often gastric adenocarcinoma. The paraneoplastic syndrome is known as the **sign of Leser-Trélat**. Eruptive seborrheic keratoses that are not associated with cancer are sometimes described as having pseudo-sign of Leser-Trélat.

Eruptive and irritated seborrheic keratoses may also arise as an adverse reaction to a medication, such as [adalimumab](#), [vemurafenib](#), [dabrafenib](#), [5-fluorouracil](#) and many [chemotherapy drugs](#).

An irritated seborrheic keratosis is an inflamed, red and crusted lesion. It may give rise to [eczematous dermatitis](#) around the seborrheic keratosis. Dermatitis may also trigger new seborrheic keratoses to appear.

How is a seborrheic keratosis diagnosed?

The diagnosis of seborrheic keratosis is often easy.

- A stuck-on, well-demarcated warty plaque
- Other similar lesions

Sometimes, seborrheic keratosis may resemble [skin cancer](#), such as [basal cell carcinoma](#), [squamous cell carcinoma](#) or [melanoma](#).

[Dermoscopy](#) often shows a disordered structure in a [seborrheic keratosis](#), as is also true for skin cancer. There are diagnostic dermatoscopic clues to seborrheic keratosis, such as multiple orange or brown clods (due to keratin in skin surface crevices), white milia-like clods, and curved thick ridges and furrows forming a brain-like or cerebriform pattern.

If doubt remains, a seborrheic keratosis may undergo partial [shave or punch biopsy](#) or diagnostic [excision](#). [see [Seborrheic keratosis pathology](#)]

The dominant histopathological features of seborrheic keratosis may be described as:

- Melanoacanthoma (deeply pigmented)
- Acanthotic
- Hyperkeratotic or papillomatous
- Adenoid or reticulated
- Clonal or nested
- Adamantinoid or mucinous
- Desmoplastic
- Irritated.

What is the treatment for seborrheic keratoses?

An individual seborrheic keratosis can easily be removed if desired. Reasons for removal may be that it is unsightly, itchy, or catches on clothing.

Methods used to remove seborrheic keratoses include:

- [Cryotherapy](#) (liquid nitrogen) for thinner lesions (repeated if necessary)
- [Curettage](#) and/or [electrocautery](#)
- [Ablative laser surgery](#)
- [Shave biopsy](#) (shaving off with a scalpel)
- Focal [chemical peel](#) with trichloroacetic acid

All methods have disadvantages. Treatment-induced loss of pigmentation is a particular issue for dark-skinned patients. There is no easy way to remove multiple lesions on a single occasion.

How can seborrhoeic keratoses be prevented?

How to prevent seborrhoeic keratoses is unknown.

What is the outlook for seborrhoeic keratoses?

Seborrhoeic keratoses tend to persist. From time to time, individual or multiple lesions may remit spontaneously or via the [lichenoid keratosis](#) mechanism.

Those associated with [dermatitis](#) may regress after it has been controlled.